

DHEEKSHANYA SENTHAMILSELVAN

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EDUCATION

Drexel University

Bachelor of Science in Computer Science
Minor in Math and Software Engineering

Philadelphia, PA
Anticipated Graduation: June 2027
GPA 3.99

Courses Taken: Data Structures & Algorithms, Computer Architecture, Discrete Math

TECHNICAL SKILLS

Languages: C++, C#, Java, TypeScript, Python, SQL, C

Frameworks & Libraries: .NET, Spring Boot, React, Dexie.js

Dev Tools: Git, Docker, GitHub Actions, Vite

Databases & Cloud: PostgreSQL, Redis, Supabase, AWS (Lambda, ECR, API Gateway), RESTful APIs

WORK EXPERIENCE

Software Developer Co-op

Susquehanna International Group

Mar 2026 – Present

Bala Cynwyd, PA

- Architected an end-to-end travel reconciliation platform using **C#, SQL, SSIS, REST APIs, and React**, automating Amex/vendor data matching and exception handling.
- Built a **35+ component React/TypeScript library with Tailwind**, theming, CI/CD, and internal package distribution, driving migration from legacy C#/Kendo UIs.
- Developed **.NET APIs and MCP/AI tooling**, integrating internal LLMs with enterprise data and **building evals**; implemented authentication, async processing, testing, logging, and CI/CD.

Data Analyst Co-op

PECO, Exelon

Mar 2025 – Sept 2025

Philadelphia, PA

- Collaborated with engineers to integrate internal Python micro-services into analytics dashboards.
- Supported early-stage development of a machine learning model for gas leak detection; shared progress at a company tech showcase.
- Used Docker to package internal tools, making setup and deployment consistent across machines.

PERSONAL PROJECTS

Tech Summit: Tech Conference App (PWA) | React, TypeScript, Vite, IndexedDB, Supabase

Dec 2025

- Engineered an offline-first React PWA with local IndexedDB storage and Supabase sync.
- Built features such as QR-based networking, session tracking, and voice notes.

SkillSwap-Hub | Java, Spring Boot, React, PostgreSQL, Redis

Oct 2025

- Built a skill-based barter platform with session booking, chat, reviews, and credit-based transactions using Java 21, Spring Boot, PostgreSQL, Redis, and WebSockets

Sliding Puzzle Solver | C, A*, Python, Docker, AWS Lambda, ECR, API Gateway, REST APIs

Mar 2025

- Developed and deployed an optimal sliding puzzle solver using A* search algorithm with Manhattan distance heuristic, achieving sub-5-second execution for puzzles up to 10x10
- Built complete CI/CD pipeline using Docker, AWS ECR, and Lambda with HTTP API Gateway integration for RESTful endpoint access

RESEARCH EXPERIENCE

Vision Language Models for Autonomous Driving

Research Assistant

Sept 2025 – Present

Drexel University

- Build python tooling to generate and manage large-scale synthetic navigation datasets in simulated environments.
- Wrote scripts to fine-tune and evaluate vision language pipelines, focusing on data preprocessing and reproducible workflows